

PSEUDOCODE FOR THE PROJECT

BEGIN

DISPLAY "=====

DISPLAY " LED RESISTOR VALUE CALCULATOR SYSTEM "

DISPLAY "=====

INPUT numberOfCircuits

INITIALIZE circuitRecords AS Array of LEDCircuit Objects

FOR each circuit FROM 1 TO numberOfCircuits DO

INPUT ledName, supplyVoltage, forwardVoltage, ledCurrentMilliAmps

// Step 5: Input Validation Diamond

IF (supplyVoltage <= forwardVoltage) OR (ledCurrentMilliAmps <= 0) THEN

// Step 6a: Invalid Path (Bypasses Math, Safety Thresholds, and Display)

SET designStatus = "Invalid Design"

ELSE

// Step 6b: Sequential Calculations (Valid Path)

SET ledCurrentAmps = ledCurrentMilliAmps / 1000

SET resistorValue = (supplyVoltage - forwardVoltage) / ledCurrentAmps

SET resistorPower = ledCurrentAmps * ledCurrentAmps * resistorValue

// Step 7 & 8: Safety Evaluation

IF (ledCurrentMilliAmps > 30) THEN

SET designStatus = "Warning: LED current may be too high for a standard LED."

ELSE

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        SET designStatus = "Design Accepted"
    END IF

    // Step 9: Print results for valid configuration
    DISPLAY resistorValue, resistorPower, designStatus
END IF

STORE current circuit details into circuitRecords

// Step 10: Decision Diamond (Loop Evaluator)
// IF NO: The FOR loop automatically branches back to Step 3
// IF YES: The loop completes naturally and proceeds down
END FOR

// Step 11: File Saving Sequence
OPEN file stream FOR "led_resistor_report.txt"
WRITE circuitRecords summary data to file
CLOSE file stream

DISPLAY "Report successfully generated!"
END
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